



Smart Stadium Staff App

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Abstract

OpsLite is a real-time coordination platform for large-scale venues like **Dragon Stadium**, bridging field staff via mobile apps and supervisors through a web dashboard.

By integrating **STM32N6** for automated crowd counting and incident detection, the system — originally designed for stadium management — was validated within the **IT Aveiro building** to ensure safety and operational efficiency.

Implementation

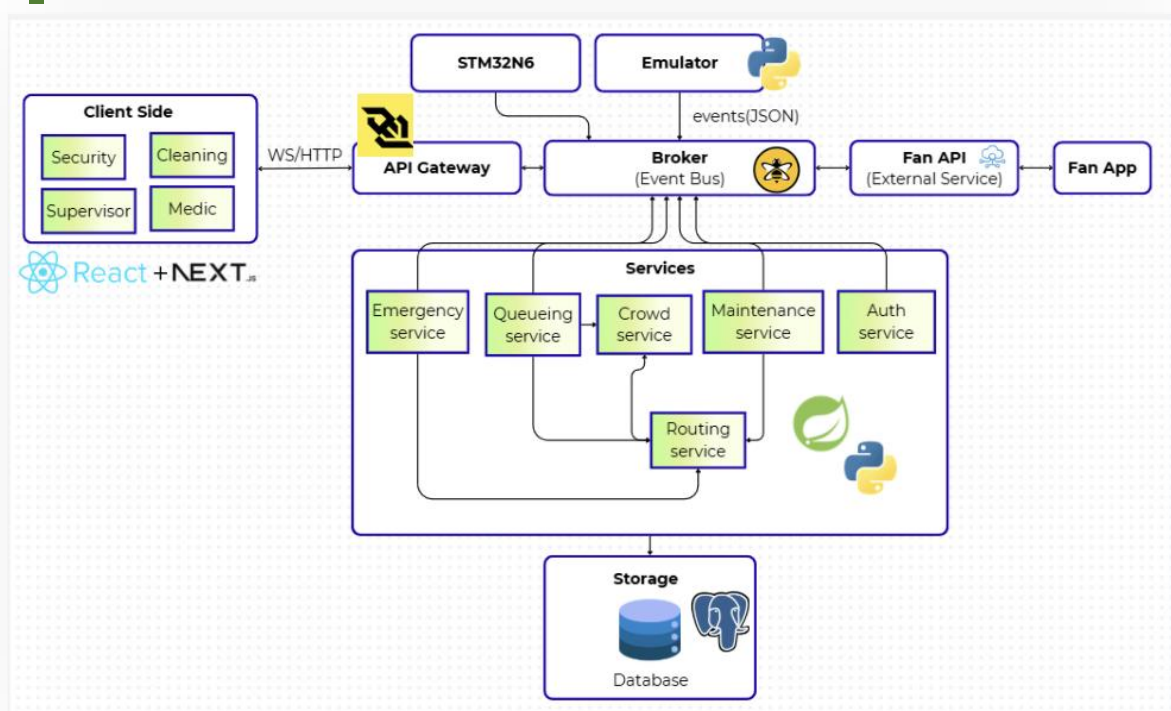


Fig 1 - System Architecture

Methodologies

- Algorithms & Modeling:

- Uses **A*** pathfinding for multi-floor routing and dynamic obstacle avoidance.
- Predicts gate wait times using **M/M/1** queueing theory.

- Artificial Intelligence:

- Executes object detection models on **STM32N6** hardware to count people at gates and detect incidents.



Fig 2 – ZIP and CNN AI models for crowd counting and people detection

Features

- **Real-Time Monitoring:** Operators view live crowd density heatmaps and receive alerts when sections reach critical occupancy thresholds. (Fig. 3).

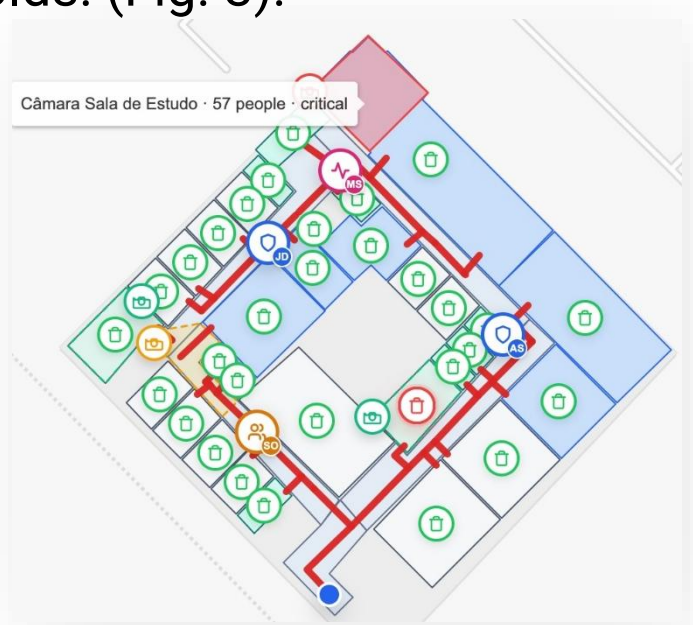


Fig 3 – Heatmaps data

- **Smart Routing:** Calculates optimal multi-floor paths while dynamically avoiding congested zones (Fig. 4).



Fig 4 - Multi-floor paths

- **SOS Coordination:** Assigns and routes the nearest responder to active emergency alerts in real time (Fig. 5).

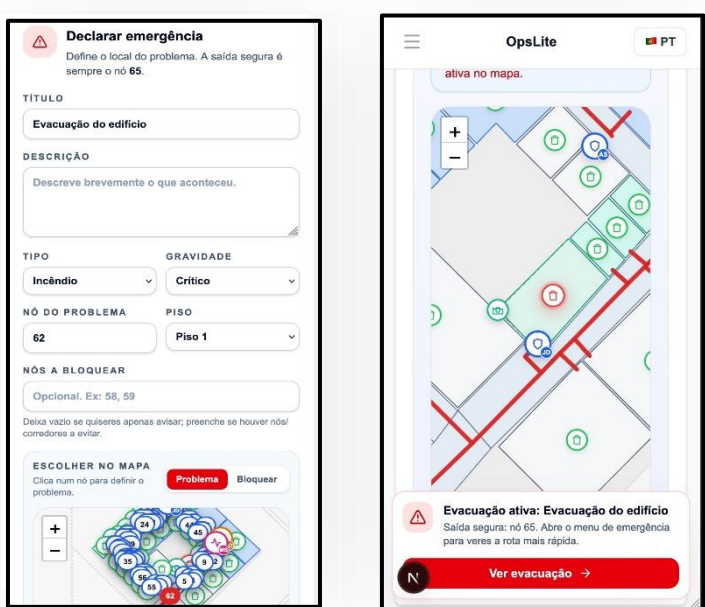


Fig 5 - SOS Coordination

- **Staff Coordination:** Location-aware task dispatching and mission-specific chat.



Fig 6 – Task management